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The Archive

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September 26, 2020

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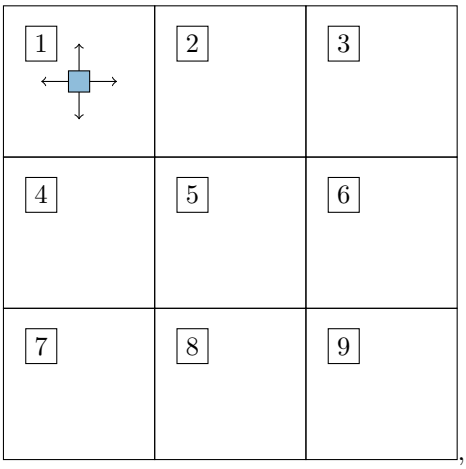
Reinforcement Learning Theory

Adapted from *Neuro-Dynamic Programming* by
Dmitri P. Bertsekas and John N. Tsitsiklis.

Physical intuition.

Consider the following "environment":



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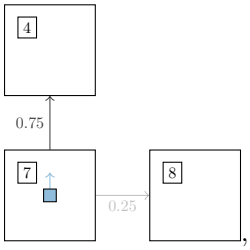
where there are a total of 9 "states" (▢), an "agent" (■), and a set of "actions" (→) that the agent can take at any state.

At every "step" in time, the agent is in one of the states, and must take one of the actions.

At any step, if the agent takes an action in a state, it will be in a state at the next step according to a "transition distribution" (that only depends on the current state and action, not the current step).

For example, the agent could

– take action ↑ at state 7:



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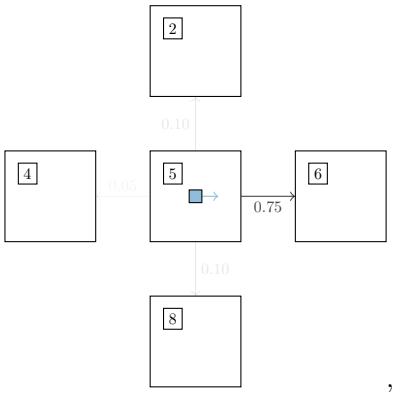
and

– take action → at state 5:



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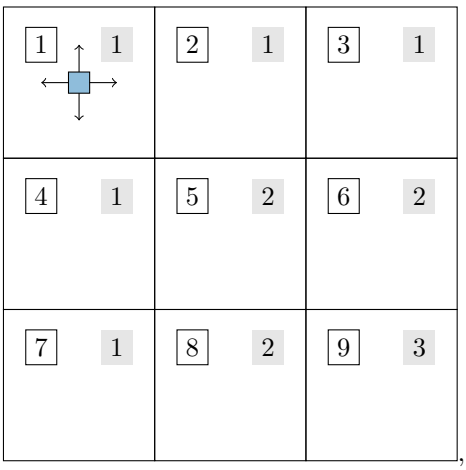
showing the conditional probability of being in another state at the next step according to the transition distribution.

However, this begs the question, what state is the agent at initially (step 1)? To decide this, there is also an initial distribution over the states.

Now, let's make things a bit more interesting.

Let every state also have a return that the agent recieves from it, leaving our environment looking something like:

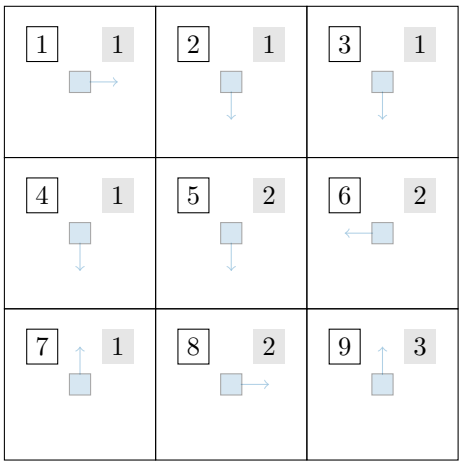


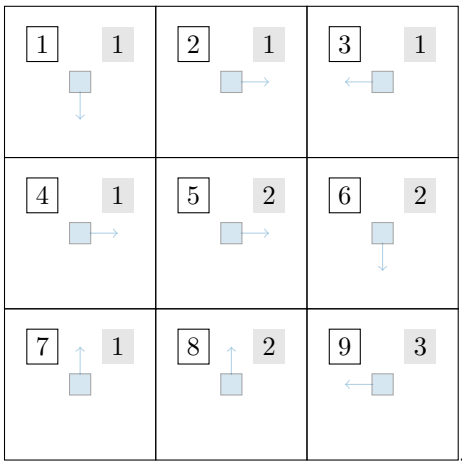
,

with the returns (·) shown.

Also, let there be some fixed "step limit" such that, when the current step reaches it, everything ends.

Now, suppose you can "program" the agent as follows: for every state and step, you can fix the action that the agent takes. For example, here's what you might program the agent to do at every state for two different steps:





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We'll call each such programming a "strategy".

We know that the agent starts in some state according to the initial distribution. After this, we follow the strategy, transitioning states according to the transition distribution, recieving returns at every state until we reach the step limit. Let's refer to this process as a "run", and the sum of the returns over all a run's states as the "total return".

Because there is some initial distribution and transition distribution that are fixed over runs, we know there must be some average total return for every strategy.

Now, consider the following question: what strategy has the greatest average total return?

Notation.

To answer this question, let's establish some notation.

[Define]$^{26}_{\mathrm{D}}$

$$\mathbf{S} \coloneqq \text{the set of states},$$

where $\mathbf{S}$ is finite and "s" denotes an element of $\mathbf{S}$.

[Define]$^{30}_{\mathrm{D}}$

$$\mathbf{A} \coloneqq \text{the set of actions},$$

where $\mathbf{A}$ is finite and "a" denotes an element of $\mathbf{A}$.

[Define]$^{19}_{\mathrm{D}}$

$$[R(\cdot) : \mathbf{S} \to \mathbb{R}]^{17}_{\mathrm{N}} \coloneqq \text{mapping from states to returns}.$$

[Define]$^{21}_{\mathrm{D}}$

$$T \coloneqq \text{the step limit, where } T \geq 1,$$

where t denotes a step, with $1 \leq t \leq T$.

[Define]$^{34}_{\mathrm{D}}$

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies},$$

where π denotes an element of $\mathbf{\Pi}$, with

$$\pi = (\pi_1(\cdot), \ldots, \pi_{T-1}(\cdot)) \coloneqq \text{a strategy, where each "step-strategy" } \pi_t(\cdot) : \mathbf{S} \to \mathbf{A}.$$

[Note that]$^{4}_{\mathrm{P}}$ the total number of strategies can be written as

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|.$$

[Define]$^{23}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}$,

$$[\mathrm{P_I}(\mathrm{s})]^{13}_{\mathrm{N}} \coloneqq \text{the initial distribution probability of starting in state s.}$$

[Define]$^{28}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$,

$$\mathrm{P_A}(\pi, t : \mathrm{a} \mid \mathrm{s}) \coloneqq \begin{cases} 1 & \pi_t(\mathrm{s}) = \mathrm{a} \\ 0 & \pi_t(\mathrm{s}) \neq \mathrm{a}. \end{cases}$$

[Define]$^{31}_{\mathrm{D}}$ for any $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$,

$$\mathrm{P_T}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}') \coloneqq \text{the transition distribution probability of going to state s after action a in state s}'.$$

[Define]$^{29}_{\mathrm{D}}$ for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{variable for the state at step } t,$$

with domain $\mathbf{S}$.

[Define]$^{27}_{\mathrm{D}}$ for all $1 \leq t \leq T-1$,

$$\mathrm{A}_t \coloneqq \text{variable for the action at step } t,$$

with domain $\mathbf{A}$.

[Let]$^{49}_{\mathrm{D}}$

$$\mathrm{S} \leftarrow \{(\mathrm{S}_1, \mathbf{S}), (\mathrm{A}_1, \mathbf{A}), (\mathrm{S}_2, \mathbf{S}), \ldots, (\mathrm{A}_{T-1}, \mathbf{A}), (\mathrm{S}_T, \mathbf{S})\},$$

and, for each $\pi \in \mathbf{\Pi}$, let $\mathrm{D}_\pi \in \mathbf{D}$ be a distribution.

[Let]$^{20}_{\mathrm{D}}$

$$\mathrm{Pr}_\pi(\cdot) \coloneqq \mathrm{Pr}(\cdot),$$

where $\mathrm{D} = \mathrm{D}_\pi$, as applied to [joint probabilities]$^{43}_{\mathrm{D}}$, [marginal probabilities]$^{46}_{\mathrm{D}}$, and [conditional probabilities]$^{48}_{\mathrm{D}}$.

Because we follow a fixed strategy, [we know that]$^{25}_{\mathrm{D}}$, for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $1 \leq t \leq T - 1$, when defined,

$$\Pr_\pi (\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_\mathrm{A} (\pi, t : \mathrm{a} \mid \mathrm{s}),$$

[from which it follows]$^{3}_{\mathrm{P}}$ that A_t is independent of all previous states and actions given S_t. Notationally, for all $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T - 1$, when defined,

$$\Pr_\pi (\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi (\mathrm{A}_t \mid \mathrm{S}_t).$$

[We know that]$^{18}_{\mathrm{D}}$ for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$,

$$\Pr_\pi (\mathrm{S}_1 = \mathrm{s}) = \mathrm{P}_\mathrm{I} (\mathrm{s}),$$

and [we know that]$^{33}_{\mathrm{D}}$ for all $\pi \in \mathbf{\Pi}$, $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $2 \leq t \leq T$, when defined,

$$\Pr_\pi (\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') = \mathrm{P}_\mathrm{T} (\mathrm{s} \mid \mathrm{a}, \mathrm{s}'),$$

[where]$^{24}_{\mathrm{D}}$ S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for all $\pi \in \mathbf{\Pi}$, $2 \leq t \leq T$, when defined,

$$\Pr_\pi (\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi (\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}).$$

Propositions.

Let's now describe some preliminary results using this notation.

One of the first questions we may ask is, does our notation completely describe the distribution of runs for a particular strategy? That is,

$$\text{for any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_T \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{A},$$

and

$$\text{for any strategy } \pi \in \mathbf{\Pi},$$

can we determine the value of

$$\mathrm{Pr}_\pi\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T\right)?$$

The answer to this question is: yes! We have all the information we need to completely describe this joint distribution. [It can be written][5]$_{\mathrm{P}}$ as

$$\begin{aligned}\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = {} & \mathrm{P}_{\mathrm{I}}\left(\mathrm{s}_1\right)\times \\ & \mathrm{P}_{\mathrm{A}}\left(\pi, 1 : \mathrm{a}_1 \mid \mathrm{s}_1\right)\mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_2 \mid \mathrm{a}_1, \mathrm{s}_1\right)\times \\ & \mathrm{P}_{\mathrm{A}}\left(\pi, 2 : \mathrm{a}_2 \mid \mathrm{s}_2\right)\mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_3 \mid \mathrm{a}_2, \mathrm{s}_2\right)\times \\ & \ldots\times \\ & \mathrm{P}_{\mathrm{A}}\left(\pi, T-1 : \mathrm{a}_{T-1} \mid \mathrm{s}_{T-1}\right)\mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_T \mid \mathrm{a}_{T-1}, \mathrm{s}_{T-1}\right),\end{aligned}$$

or, in more compact [product notation][14]$_{\mathrm{N}}$,

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_{\mathrm{I}}\left(\mathrm{s}_1\right) \times \prod_{t=1}^{T-1} \mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right)\mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

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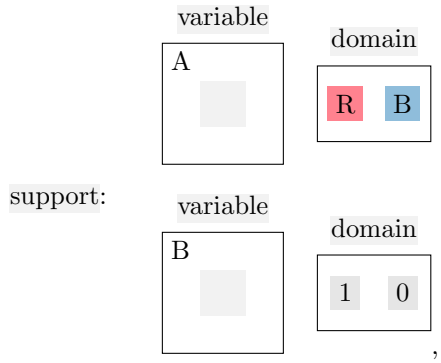
Probability Theory

Physical intuition.

Suppose that we have a set of "variables," each with their own finite "domains", which we'll collectively refer to as the "support":



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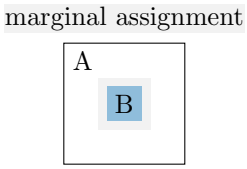
where a "marginal assignment" consists of associating some (but not all) of the variables of the support with something from their domains:



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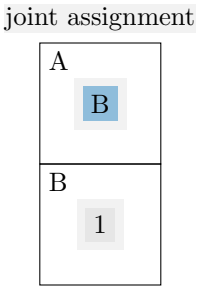
a "joint assignment" ("assignment" for short) consists of associating each variable of the support with something from its domain:



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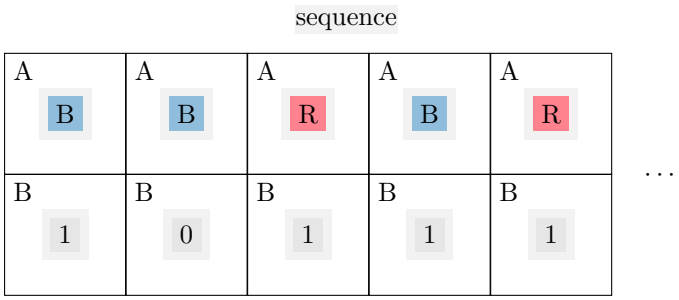
a "sequence" consists of an infinite, ordered list of such joint assignments:



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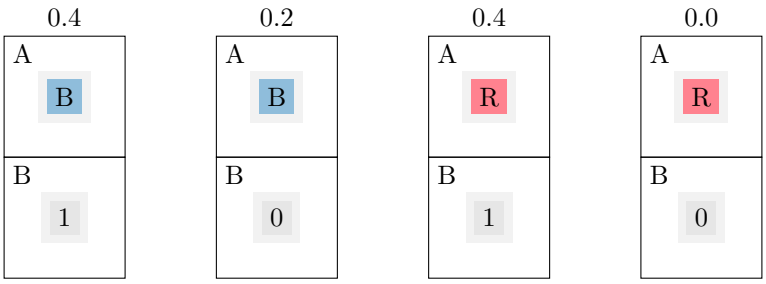
and a "trial" denotes a particular item (i.e. an indexed joint assignment) of this sequence.

Now, suppose something in particular about this sequence: if an joint assignment occurs at any trial, it will occur in infinitely many trials, and, additionally, as we the traverse sequence in order, the ratio of occurrences of the joint assignment to trials so far becomes arbitrarily close to some positive number. Otherwise, if an joint assignment never occurs, the ratio remains at zero. Let's call this ratio the "sequence probability" of the joint assignment. Let's refer to such a sequence as a probability sequence.

For example, if the sequence above repeated the first five trials, it would be a probability sequence defining the sequence probabilities:

sequence probabilities



Let a "distribution" be a set of probability sequences with the same sequence probability for every assignment, which, in the context of a distribution, we will simply refer to as the "joint probability" ("probability" for short) of that assignment.

Notation.

Let's correspond this intuition with some notation, and use this notation to arrive at a few more definitions.

[Define]$^{36}_{\mathrm{D}}$

$$\mathrm{X} \coloneqq \text{an arbitrary variable with domain } \mathbf{X},$$

where we use "x" to denote an element of $\mathbf{X}$.

We implicitly maintain that anything we describe using X and $\mathbf{X}$ also applies when these are replaced with one or more variables/domains: $\mathrm{X}/\mathbf{X}$, ..., $\mathrm{X}'/\mathbf{X}'$. Notational expansions are described where applicable.

Define $\mathrm{Y}, \mathbf{Y}, \mathrm{y}$, and $\mathrm{Z}, \mathbf{Z}, \mathrm{z}$ similarly.

Define $?, \mathbf{?}, ?$ similarly, but instead implicitly maintain that anything we describe using ? and **?** also applies when these are replaced with *zero* or more variables/domains.

[We let the notation]$^{37}_{\mathrm{D}}$

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\} \text{ establish a support consisting of variable } \mathrm{X} \text{ with domain } \mathbf{X},$$

which expands as

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}.$$

[Let]$^{44}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ and all $\mathrm{x} \in \mathbf{X}$,

$$\mathrm{X} = \mathrm{x} \text{ denote a joint assignment (assignment for short)},$$

which for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}$ and all $\mathrm{x} \in \mathbf{X}$, ..., $\mathrm{x}' \in \mathbf{X}'$, expands as

$$\mathrm{X} = \mathrm{x}, \ldots, \mathrm{X}' = \mathrm{x}'.$$

[Let]$^{45}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ and all $\mathrm{x} \in \mathbf{X}$,

$$\mathrm{X} = \mathrm{x} \text{ denote a marginal assignment},$$

which for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}'), (\mathrm{Y}, \mathbf{Y}), \ldots, (\mathrm{Y}', \mathbf{Y}')\}$ and all $\mathrm{x} \in \mathbf{X}$, ..., $\mathrm{x}' \in \mathbf{X}'$, expands as

$$\mathrm{X} = \mathrm{x}, \ldots, \mathrm{X}' = \mathrm{x}'.$$

In simple English, a list of equalities that doesn't cover the entire support is a marginal assignment.

[Let]$^{38}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQ} \text{ denote the set of all sequences},$$

where "sq" denotes an arbitrary such sequence.

Let $1 \leq t < \infty$ denote an arbitrary trial.

Because sequences are infinite (and cannot be written out), we use the notation

$$\mathrm{SQI}\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) \coloneqq \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$.

[Define]$^{40}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\,(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \mathrm{SQI}\,(\mathrm{sq}, t', \mathrm{X} = \mathrm{x})}{t} \right)$$

if such a limit exists, and undefined otherwise; that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[Let]$^{41}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}, (\exists\ 1 \leq t < \infty : \mathrm{SQI}\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1) \implies \Pr\,(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0\},$$

that is, the set of all probability sequences, where "sqp" denotes an arbitrary such probability sequence.

[Define]$^{42}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\ \mathrm{d}, \mathrm{d}' \in \mathrm{D}, \forall\ \mathrm{x} \in \mathbf{X}, \Pr\,(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions, where "D" denotes an abitrary such distribution, and "d" denotes an arbitrary sequence in D.

[Define]$^{43}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$ and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\,(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr\,(\mathrm{X} = \mathrm{x}) = \Pr\,(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a joint probability.

[Define]$^{46}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\,(\mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr\,(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}),$$

that is, a marginal probability.

[It can be shown]$^{47}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\,(\mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr\,(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[Define]$^{48}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$,

$$\Pr\,(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \frac{\Pr\,(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr\,(\mathrm{Y} = \mathrm{y})},$$

and undefined when $\Pr\,(\mathrm{Y} = \mathrm{y}) = 0$; that is, a conditional probability.

Propositions.

[It can be shown that]$^{50}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}) = 0 \implies \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

[It can be shown that]$^{51}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}) > 0.$$

[It can be shown that]$^{52}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$, and $\mathrm{z} \in \mathbf{Z}$, when $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, and $\Pr(\mathrm{Y} = \mathrm{y}) > 0$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1 \implies \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 1.$$

[It can be shown that]$^{53}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$, and $\mathrm{z} \in \mathbf{Z}$, when $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, and $\Pr(\mathrm{Y} = \mathrm{y}) > 0$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 0 \implies \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0.$$

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[3] /T/[rl_theory]$^{2}_{T}$/P/[actcind]$^{3}_{P}$

Proposition.

We have that A_t is independent of all previous states and actions given S_t. Notationally, for all $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$, when defined,

$$\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) = \Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right).$$

Proof.

Because $\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right)$ is defined, we have that

$$\Pr_\pi\left(\mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) > 0,$$

implying that

$$\Pr_\pi\left(\mathrm{S}_t\right) > 0.$$

Therefore, $\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right)$ is also defined. Suppose $\mathrm{A}_t = \mathrm{a} \in \mathbf{A}$ and $\mathrm{S}_t = \mathrm{s} \in \mathbf{S}$. [It follows by definition]$^{25}_{D}$ that

$$\Pr_\pi\left(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}\right) = \mathrm{P}_\mathrm{A}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right),$$

so [there are two cases]$^{28}_{D}$ to consider:

- Case 1: $\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right) = 1$. Then, po

$$\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) \overset{\text{(po)->[...]}^{52}_{P}\ \text{po}}{=} 1 = \Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right).$$

- Case 2: $\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right) = 0$. Then, pz

$$\Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) \overset{\text{(pz)->[...]}^{53}_{P}\ \text{pz}}{=} 0 = \Pr_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right).$$

[4] /T/[rl_theory]$^{2}_{T}$/P/[npol]$^{4}_{P}$

Proposition.

We have that the total number of strategies can be written as

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|.$$

Proof.

This follows from simple combinatorics. Consider an arbitrary strategy

$$\pi = (\pi_1(\cdot), \ldots, \pi_T(\cdot)).$$

Because each step-strategy $\pi_t(\cdot) : \mathbf{S} \to \mathbf{A}$, there are a total of $|\mathbf{S}| \times |\mathbf{A}|$ distinct $\pi_t(\cdot)$ possible.

Because each strategy $\pi \in \mathbf{\Pi}$ consists of T such step-strategies $\pi_t(\cdot)$, there are a total of

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|$$

distinct π possible.

[5] /T/[rl_theory]$^{2}_{T}$/P/[pexpr]$^{5}_{P}$

Proposition.

We have that for

$$\text{any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_T \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T\right)$$

as

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t=1}^{T-1} \mathrm{P}_\mathrm{A}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

Proof.

[It can be shown inductively]$^{11}_{P}$ that for any $1 \leq t \leq T$, for

$$\text{any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_t \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t\right)$$

as

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right),$$

for which the result follows in particular when $t = T$.

[13] /T/[rl_theory]$^{2}_{T}$/N/[paramprob]$^{13}_{N}$

Note.

We use the notation $\Pr(\cdot)$ and $\mathrm{P}(\cdot)$ to distinguish between a probability and a "probability parameterization", respectively.

A probability parameterization is simply a function mapping to probability values, and is defined for every possible input. The form of the inputs are implicit in its definition.

A probability, however, may or may not be defined, specifically when it is a conditional probability.

When we correspond a probability with a probability parameterization, we mean to say that, for a particular input, they are equal when the probability is defined.

For example, suppose $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ with any D.

For all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$, if we correspond $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ with $\mathrm{P}(\mathrm{x} \mid \mathrm{y})$, we mean to say that when $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ is defined (that is, when $\Pr(\mathrm{Y} = \mathrm{y}) > 0$),

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \mathrm{P}(\mathrm{x} \mid \mathrm{y}).$$

[14] /T/[rl_theory]$^{2}_{T}$/N/[prodnot]$^{14}_{N}$

Note.

Consider an arbitrary function $f(\cdot)$. The notation

$$\prod_{x=a}^{b} f(x),$$

where a, b are integers, denotes the product $f(a) \times f(a+1) \times \cdots \times f(b-1) \times f(b)$ when $a \leq b$, and 1 otherwise. That is,

$$\prod_{x=a}^{b} f(x) = \begin{cases} f(a) \times f(a+1) \times \cdots \times f(b-1) \times f(b) & a \leq b \\ 1 & a > b. \end{cases}$$

[17] /T/[rl_theory]$^{2}_{T}$/N/[fnnot]$^{17}_{N}$

Note.

We use the notation

$$f(\cdot) : X \to Y$$

to denote a function with domain X and range Y; that is,

$$\text{for all } x \in X, \text{ there is some } y \in Y \text{ such that } f(x) = y.$$

[18] /T/[rl_theory]$^{2}_{T}$/D/[ind]$^{18}_{D}$

Definition.

We have by definition that for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$,

$$\mathrm{Pr}_{\pi}\left(\mathrm{S}_1 = \mathrm{s}\right) = \mathrm{P}_{\mathrm{I}}\left(\mathrm{s}\right).$$

[19] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[rewards]$^{19}_{\mathrm{D}}$

Definition.

We have by definition that

$$R(\cdot) : \mathbf{S} \to \mathbb{R} \coloneqq \text{mapping from states to returns.}$$

[20] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[pstr]$^{20}_{\mathrm{D}}$

Definition.

We have by definition that

$$\Pr_{\pi}(\cdot) \coloneqq \Pr(\cdot),$$

where $\mathrm{D} = \mathrm{D}_{\pi}$, as applied to [joint probabilities]$^{43}_{\mathrm{D}}$, [marginal probabilities]$^{46}_{\mathrm{D}}$, and [conditional probabilities]$^{48}_{\mathrm{D}}$.

[21] /T/[rl_theory]$^{2}_{T}$/D/[steplim]$^{21}_{D}$

Definition.

We have by definition that

$$T \coloneqq \text{the step limit, where } T \geq 1,$$

where t denotes a step, with $1 \leq t \leq T$.

[23] /T/[rl_theory]$^{2}_{T}$/D/[inddef]$^{23}_{D}$

Definition.

We have by definition that for any $s \in \mathbf{S}$,

$$P_I(s) \coloneqq \text{the initial distribution probability of starting in state } s.$$

[24] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[trdcind]$^{24}_{\mathrm{D}}$

Definition.

We have by definition that S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for all $\pi \in \mathbf{\Pi}$, $2 \leq t \leq T$, when defined,

$$\mathrm{Pr}_\pi\left(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) = \mathrm{Pr}_\pi\left(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}\right).$$

[25] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[pact]$^{25}_{\mathrm{D}}$

Definition.

We have by definition that, for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $1 \leq t \leq T - 1$, when defined,

$$\mathrm{Pr}_{\pi}\left(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}\right) = \mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right).$$

[26] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[states]$^{26}_{\mathrm{D}}$

Definition.

We have by definition that

$$\mathbf{S} := \text{the set of states},$$

where $\mathbf{S}$ is finite and "s" denotes an element of $\mathbf{S}$.

[27] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[rvat]$^{27}_{\mathrm{D}}$

Definition.

We have by definition that for all $1 \leq t \leq T - 1$,

$$\mathrm{A}_t \coloneqq \text{variable for the action at step } t,$$

with domain $\mathbf{A}$.

[28] /T/[rl_theory]$^{2}_{T}$/D/[pactdef]$^{28}_{D}$

Definition.

We have by definition that for any $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$,

$$\mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right) \coloneqq \begin{cases} 1 & \pi_t\left(\mathrm{s}\right) = \mathrm{a} \\ 0 & \pi_t\left(\mathrm{s}\right) \neq \mathrm{a}. \end{cases}$$

[29] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[rvst]$^{29}_{\mathrm{D}}$

Definition.

We have by definition that for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{variable for the state at step } t,$$

with domain $\mathbf{S}$.

[30] /T/[rl_theory]$^{2}_{T}$/D/[actions]$^{30}_{D}$

Definition.

We have by definition that

$$\mathbf{A} := \text{the set of actions},$$

where $\mathbf{A}$ is finite and "a" denotes an element of $\mathbf{A}$.

[49] /T/[rl_theory]$^{2}_{T}$/D/[psp]$^{49}_{D}$

Definition.

We have by definition that

$$\mathrm{S} \leftarrow \{(\mathrm{S}_1, \mathbf{S}), (\mathrm{A}_1, \mathbf{A}), (\mathrm{S}_2, \mathbf{S}), \ldots, (\mathrm{A}_{T-1}, \mathbf{A}), (\mathrm{S}_T, \mathbf{S})\},$$

and, for each $\pi \in \mathbf{\Pi}$, let $\mathrm{D}_\pi \in \mathbf{D}$ be a distribution.

[31] /T/[rl_theory]$^{2}_{T}$/D/[trddef]$^{31}_{D}$

Definition.

We have by definition that for any $s, s' \in \mathbf{S}$, $a \in \mathbf{A}$,

$P_T(s \mid a, s') \coloneqq$ the transition distribution probability of going to state s after action a in state s'.

[33] /T/[rl_theory]$^{2}_{T}$/D/[trd]$^{33}_{D}$

Definition.

We have by definition that for all $\pi \in \mathbf{\Pi}$, $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $2 \leq t \leq T$, when defined,

$$\mathrm{Pr}_{\pi}\left(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}'\right) = \mathrm{P}_{\mathrm{T}}\left(\mathrm{s} \mid \mathrm{a}, \mathrm{s}'\right).$$

[34] /T/[rl_theory]$^{2}_{\mathrm{T}}$/D/[strs]$^{34}_{\mathrm{D}}$

Definition.

We have by definition that

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies,}$$

where π denotes an element of $\mathbf{\Pi}$, with

$$\pi = (\pi_1(\cdot), \ldots, \pi_{T-1}(\cdot)) \coloneqq \text{a strategy, where each “step-strategy” } \pi_t(\cdot) : \mathbf{S} \to \mathbf{A}.$$

[53] /T/[prob]$^{35}_{T}$/P/[pzerocind]$^{53}_{P}$

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, and all $x \in \mathbf{X}$, $y \in \mathbf{Y}$, and $z \in \mathbf{Z}$, when $\Pr(Y = y, Z = z) > 0$, and $\Pr(Y = y) > 0$,

$$\overset{\texttt{mg}}{\Pr(X = x \mid Y = y) = 0} \overset{\texttt{mmg}}{\implies} \Pr(X = x \mid Y = y, Z = z) = 0.$$

Proof.

We have that

$$\Pr(x \mid y) = \frac{\Pr(x, y)}{\Pr(y)} \overset{\texttt{(mmg)->[...]}^{48}_{D}}{=} \overset{\texttt{cz}}{0},$$

so

$$\Pr(x, y) = \overset{\texttt{mz}}{0},$$

and we have

$$\Pr(x, y, z) = \overset{\texttt{(mz)->[...]}^{50}_{P}}{0},$$

so

$$\Pr(x \mid y, z) = \frac{\Pr(x, y, z)}{\Pr(y, z)} \overset{\texttt{(mg)->[...]}^{48}_{D}}{=} \overset{\texttt{fz}}{0}.$$

[52] /T/[prob]$^{35}_{T}$/P/[ponecind]$^{52}_{P}$

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, and all $x \in \mathbf{X}$, $y \in \mathbf{Y}$, and $z \in \mathbf{Z}$, when $\Pr(Y = y, Z = z) > 0$, and $\Pr(Y = y) > 0$,

$$\overset{\texttt{mg}}{\Pr(X = x \mid Y = y) = 1} \overset{\texttt{mmg}}{\implies} \Pr(X = x \mid Y = y, Z = z) = 1.$$

Proof.

We have that

$$\Pr(x \mid y) = \frac{\Pr(x, y)}{\Pr(y)} \overset{\texttt{(mmg)->}[\ldots]^{48}_{D}\ \texttt{co}}{=} 1,$$

so

$$\Pr(x, y) \underset{\texttt{me}}{=} \Pr(y) \overset{[\ldots]^{47}_{P}}{=} \Pr(x, y) + \sum_{x' \in \mathbf{X}, x' \neq x} \Pr(x', y),$$

and we have

$$\forall x' \in \mathbf{X}, x' \neq x, \Pr(x', y) = 0,$$

so

$$\forall x' \in \mathbf{X}, x' \neq x, \Pr(x', y, z) \underset{\texttt{az}}{\overset{[\ldots]^{50}_{P}}{=}} 0,$$

and Therefore,

$$\Pr(x, y, z) \underset{\texttt{arm}}{\overset{\texttt{az}}{=}} \Pr(x, y, z) + \sum_{x' \in \mathbf{X}, x' \neq x} \Pr(x', y, z) \underset{\texttt{rm}}{\overset{[\ldots]^{47}_{P}}{=}} \Pr(y, z) \underset{\texttt{gf}}{\overset{\texttt{mg}}{>}} 0,$$

so

$$\Pr(x \mid y, z) = \frac{\Pr(x, y, z)}{\Pr(y, z)} \overset{\texttt{(mg)->}[\ldots]^{48}_{D}\ \texttt{arm-rm,gf}}{=} 1.$$

[50] /T/[prob]$^{35}_{T}$/P/[margzimpl]$^{50}_{P}$

Proposition.

We have that for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, and all $x \in \mathbf{X}$, $y \in \mathbf{Y}$,

$$\Pr(X = x) = 0 \overset{\texttt{mz}}{\implies} \Pr(X = x, Y = y) = 0.$$

Proof.

We have that

$$0 \overset{\texttt{mz}}{=} \Pr(X = x) \overset{[\dots]^{47}_{P}}{=} \sum_{y' \in \mathbf{Y}} \Pr(X = x, Y = y'),$$

so in particular

$$\Pr(X = x, Y = y) = 0.$$

[47] /T/prob[35]T/P/mprobg[47]P

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

Proof.

We have by definition that for all $\mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{46}_{\mathrm{D}}}{=} \sum_{?\in\mathbf{?}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?),$$

mm

so

$$\Pr(\mathrm{X} = \mathrm{x}) \overset{[\ldots]^{46}_{\mathrm{D}}}{=} \sum_{\mathrm{y}\in\mathbf{Y}} \sum_{?\in\mathbf{?}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?) \overset{\mathtt{mm}}{=} \sum_{\mathrm{y}\in\mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[51] /T/[prob]$^{35}_{T}$/P/[marggimpl]$^{51}_{P}$

Proposition.

We have that for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, and all $x \in \mathbf{X}$, $y \in \mathbf{Y}$,

$$\Pr(X = x, Y = y) > 0 \overset{\texttt{mg}}{\implies} \Pr(X = x) > 0.$$

Proof.

We have that

$$\Pr(X = x) \overset{[\cdots]^{47}_{P}}{=} \sum_{y' \in \mathbf{Y}} \Pr(X = x, Y = y') \geq \Pr(X = x, Y = y) \overset{\texttt{mg}}{>} 0.$$

[41] /T/[prob]$^{35}_{T}$/D/[seqp]$^{41}_{D}$

Definition.

We let for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}, (\exists\ 1 \leq t < \infty : \mathrm{SQI}\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1) \implies \Pr\,(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0\},$$

that is, the set of all probability sequences, where "sqp" denotes an arbitrary such probability sequence.

[46] /T/[prob]$^{35}_{T}$/D/[mprob]$^{46}_{D}$

Definition.

We have by definition that for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y})\}$, $D \in \mathbf{D}$, and all $x \in \mathbf{X}$,

$$\Pr(X = x) = \sum_{y \in \mathbf{Y}} \Pr(X = x, Y = y),$$

that is, a marginal probability.

[42] /T/[prob]$^{35}_{T}$/D/[dist]$^{42}_{D}$

Definition.

We define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\ \mathrm{d}, \mathrm{d}' \in \mathrm{D}, \forall\ \mathrm{x} \in \mathbf{X}, \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions, where "D" denotes an abitrary such distribution, and "d" denotes an arbitrary sequence in D.

[45] /T/[prob]$^{35}_{T}$/D/[masn]$^{45}_{D}$

Definition.

We let for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y})\}$ and all $x \in \mathbf{X}$,

$$X = x \text{ denote a marginal assignment,}$$

which for contextual $S \leftarrow \{(X, \mathbf{X}), \ldots, (X', \mathbf{X}'), (Y, \mathbf{Y}), \ldots, (Y', \mathbf{Y}')\}$ and all $x \in \mathbf{X}$, $\ldots$, $x' \in \mathbf{X}'$, expands as

$$X = x, \ldots, X' = x'.$$

In simple English, a list of equalities that doesn't cover the entire support is a marginal assignment.

[43] /T/[prob]$^{35}_{T}$/D/[prob]$^{43}_{D}$

Definition.

We define for contextual $S \leftarrow \{(X, \mathbf{X})\}$, $D \in \mathbf{D}$ and all $x \in \mathbf{X}$,

$$\Pr(X = x)$$

such that

$$\forall d \in D, \Pr(X = x) = \Pr(d; X = x),$$

that is, a joint probability.

[38] /T/[prob]$^{35}_{T}$/D/[seq]$^{38}_{D}$

Definition.

We let, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQ} \text{ denote the set of all sequences,}$$

where "sq" denotes an arbitrary such sequence.

Let $1 \leq t < \infty$ denote an arbitrary trial.

Because sequences are infinite (and cannot be written out), we use the notation

$$\mathrm{SQI}\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) \coloneqq \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$.

[40] /T/[prob]$^{35}_{T}$/D/[prsq]$^{40}_{D}$

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \mathrm{SQI}(\mathrm{sq}, t', \mathrm{X} = \mathrm{x})}{t} \right)$$

if such a limit exists, and undefined otherwise; that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[44] /T/[prob]$^{35}_{T}$/D/[jasn]$^{44}_{D}$

Definition.

We let for contextual $S \leftarrow \{(X, \mathbf{X})\}$ and all $x \in \mathbf{X}$,

$$X = x \text{ denote a joint assignment (assignment for short),}$$

which for contextual $S \leftarrow \{(X, \mathbf{X}), \ldots, (X', \mathbf{X}')\}$ and all $x \in \mathbf{X}$, $\ldots$, $x' \in \mathbf{X}'$, expands as

$$X = x, \ldots, X' = x'.$$

[36] /T/[prob]$^{35}_{T}$/D/[xyz]$^{36}_{D}$

Definition.

We have by definition that

$$\mathrm{X} \coloneqq \text{an arbitrary variable with domain } \mathbf{X},$$

where we use "x" to denote an element of $\mathbf{X}$.

We implicitly maintain that anything we describe using X and $\mathbf{X}$ also applies when these are replaced with one or more variables/domains: $\mathrm{X}/\mathbf{X}$, ..., $\mathrm{X}'/\mathbf{X}'$. Notational expansions are described where applicable.

Define $\mathrm{Y}, \mathbf{Y}, \mathrm{y}$, and $\mathrm{Z}, \mathbf{Z}, \mathrm{z}$ similarly.

Define $?, \mathbf{?}, ?$ similarly, but instead implicitly maintain that anything we describe using ? and **?** also applies when these are replaced with *zero* or more variables/domains.

[37] /T/[prob]$^{35}_{T}$/D/[spt]$^{37}_{D}$

Definition.

We let the notation

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\} \text{ establish a support consisting of variable } \mathrm{X} \text{ with domain } \mathbf{X},$$

which expands as

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}.$$

[48] /T/[prob]$^{35}_{T}$/D/[cprob]$^{48}_{D}$

Definition.

We have by definition that for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, and all $x \in \mathbf{X}$, $y \in \mathbf{Y}$,

$$\Pr(X = x \mid Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)},$$

and undefined when $\Pr(Y = y) = 0$; that is, a conditional probability.

[11] /T/[rl_theory]$^{2}_{T}$/P/[pexpr]$^{5}_{P}$/P/[pexprt]$^{11}_{P}$

Proposition.

We have that for any $1 \leq t \leq T$, for

$$\text{any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_t \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t\right)$$

as

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Proof.

Base case. When $t = 1$, we have [by definition]$^{18}_{D}$ that $\mathrm{Pr}_\pi\left(\mathrm{s}_1\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right)$.

Inductive hypothesis. Assume that for $t - 1$ such that $1 \underset{\texttt{tmg}}{\leq} t - 1 \underset{\texttt{tml}}{\leq} T - 1$, the proposition holds.

That is,

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \underset{\texttt{ih}}{=} \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-2} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Inductive step. We want to show that the proposition holds for t.

There are two cases to consider:

- Case 1: $\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \underset{\texttt{co}}{=} 0$. Then,

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) \overset{\texttt{(co)->}[\ldots]^{50}_{P}}{\underset{\texttt{coz}}{=}} 0,$$

  and

$$\begin{aligned}
&\mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \\
&= \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-2} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \times \\
&\quad \mathrm{P}_\mathrm{A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
&\underset{\texttt{ih}}{=} \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
&\underset{\texttt{co}}{=} 0 \underset{\texttt{coz}}{=} \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right).
\end{aligned}$$

– Case 2: $\Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) > 0$. Then,
ct

$$\overset{\texttt{(ct)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \text{ is defined}}, \overset{\texttt{((ct)->}[\ldots]^{51}_{\mathrm{P}}\texttt{)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(a_{t-1} \mid s_{t-1}) \text{ is defined}},$$
daf da

and we have

$$\overset{\texttt{(daf,tmg,tml)->}[\ldots]^{3}_{\mathrm{P}}}{\Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) = \Pr_\pi(a_{t-1} \mid s_{t-1})} \overset{\texttt{(da,tmg,tml)->}[\ldots]^{25}_{\mathrm{D}}}{= \mathrm{P_A}(\pi, t-1 : a_{t-1} \mid s_{t-1})}.$$
cia ad

[There are two cases]$^{28}_{\mathrm{D}}$ to consider:

– Case 2.1: $\mathrm{P_A}(\pi, t-1 : a_{t-1} \mid s_{t-1}) = 1$. Then,
cto

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}, a_{t-1}) &\overset{\texttt{daf}}{\underset{\texttt{md}}{=}} \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia-ad}}{=} \mathrm{P_A}(\pi, t-1 : a_{t-1} \mid s_{t-1}) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cto}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{ct}}{\underset{\texttt{mg}}{>}} 0.
\end{aligned}$$

Therefore,

$$\overset{\texttt{(md-mg)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \text{ is defined}}, \overset{\texttt{((md-mg)->}[\ldots]^{51}_{\mathrm{P}}\texttt{)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \text{ is defined}}$$
dsf ds

and we have

$$\overset{\texttt{(dsf,tmg,tml)->}[\ldots]^{24}_{\mathrm{D}}}{\Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) = \Pr_\pi(s_t \mid a_{t-1}, s_{t-1})} \overset{\texttt{(ds,tmg,tml)->}[\ldots]^{33}_{\mathrm{D}}}{= \mathrm{P_T}(s_t \mid a_{t-1}, s_{t-1})}.$$
cis sd

Therefore,

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_t) &\overset{\texttt{daf,dsf}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \\
&\qquad \Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia,cis}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(a_{t-1} \mid s_{t-1}) \times \Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \\
&\overset{\texttt{ih,ad,sd}}{=} \mathrm{P_I}(s_1) \times \prod_{t'=1}^{t-1} \mathrm{P_A}(\pi, t' : a_{t'} \mid s_{t'}) \mathrm{P_T}(s_{t'+1} \mid a_{t'}, s_{t'}).
\end{aligned}$$

– Case 2.2: $\mathrm{P_A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) = 0$. Then,

ctt

daf

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}, \mathrm{a}_{t-1}\right) = \mathrm{Pr}_\pi\left(\mathrm{a}_{t-1} \mid \mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \times \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right)$$

afd

cia-ad

$$= \mathrm{P_A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right)$$

ctt

$$= 0.$$

afz

Therefore,

(afd-afz)->[...]$^{50}_{\mathrm{P}}$

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = 0,$$

cttz

so

cttz ctt

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = 0 = \mathrm{P_I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$